

Matteo Esposito Marroccella

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EXPERIENCE

Siemens AG - Cybersecurity & Trust

Sep. 2024 – Mar. 2026

Software Engineer (Working Student)

Munich, Germany

- Engineered a full-stack web platform using Django and Vue.js to centralize and visualize vulnerability scan results.
- Developed robust data and ingestion pipelines for the processing of security findings.
- Maintained automated deployment and CI/CD pipelines to transition the prototype into a scalable solution.

ATS Europe Srl

Dec. 2022 – Aug. 2024

Software Engineer

Verona, Italy

- Designed and developed a comprehensive web-based ERP platform from scratch, integrating CRM functionality and workflow management.
- Refactored legacy code into modern architectures and introduced new features, accelerating core workflow processing speeds by 3x while improving data security.
- Collaborated closely with C-levels and external clients to gather requirements and deploy customized software solutions.

PROJECTS

DocuQuery RAG Agent | github.com/matteoespo/docuquery-rag-agent

2026

- Engineered a local, multi-modal Agentic RAG using LangGraph, featuring semantic query routing, CRAG for hallucination control and web-search fallback.
- Adaptive long-term memory system using a sliding window combined with LLM-driven conversation summarization.
- A pipeline using PyMuPDF and a vision LLM to parse tabular data into Markdown and transform complex diagrams into searchable vector embeddings within ChromaDB.
- Architected and containerized infrastructure, linking a FastAPI backend to a Next.js frontend.

Software Simulation Pipeline | *Master's Thesis*

2025

- Engineered a Python-based procedural generation pipeline to automate the creation of varied simulation environments.
- Automated testing to execute complex scenarios, extracting and processing objective metrics to evaluate software stack robustness and behavior.
- Analyzed resulting simulation datasets to quantify system performance.

EDUCATION

Technische Universität München (TUM)

Feb. 2026

MSc in Informatics

Munich, Germany

University of Verona

Oct. 2022

BSc in Computer Science

Verona, Italy

SKILLS

Languages : Python, JavaScript, SQL

AI & LLM Integration : LangChain, LangGraph, LangSmith, ChromaDB, HuggingFace, Ollama

Cloud & DevOps : Docker, Kubernetes, Git, CI/CD, Rancher, Argo CD, Linux, Cloud-native architectures

Backend & Frameworks : FastAPI, Django, Vue.js, REST APIs, PostgreSQL, Redis, MySQL

Spoken Languages : English (C1), German (B1, actively improving), Italian (Native)